

Bernoulli numbers and the zeta function

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1 Introduction

Suppose you wanted to know the sum of the first 1000 squares. You could sit there all day adding them up, but perhaps you're familiar with the nice, simple formula:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)}{2},$$

allowing us to immediately conclude that the answer is $500 \cdot 1001 = 500500$. But what if you were still curious, wondering what the sum of the first 1000 cubes, or maybe 10th powers, is—and you don't have all day to sit around adding them up? Perhaps there's a similar formula! We'll see shortly. . .

Maybe you're not a fan of squares, and in fact you only like *squarefree* integers, that is, those not divided by any square besides 1 (or equivalently, those with prime factorization $n = p_1^{e_1} \cdots p_k^{e_k}$ with $e_i < 2$ for all i). What's the likelihood that any random integer you come across will be squarefree? That is, what's the density of squarefree integers in $\mathbb{Z}$? It turns out that we can give a nice, clean answer to this question, and also for even exponents greater than 2. . .

The solutions to these problems are powered by two beautiful pieces of technology:

1. Bernoulli numbers and
2. the zeta function $\zeta(s)$.

We will develop the basics of these technologies, then show off their combined power, using the problems posed above as punching bags.

Essentially all the following content is from Chapter 15, section 1 of the book *A Classical Introduction to Modern Number Theory* by Kenneth Ireland and Michael Rosen.

2 Bernoulli numbers

Let's begin with the first problem: the sums we are interested in are $S_m(n) := 1^m + 2^m + \cdots + n^m$ (for $n \geq 1, m \geq 0$ integers). Classically, we know

$$S_0(n) = n, \quad S_1(n) = \frac{n(n+1)}{2}, \quad S_2 = \frac{n(n+1)(2n+1)}{6}.$$

From the above, and in line with a natural geometric interpretation of these sums as volumes of stacks of cubes approximating a simplex in $\mathbb{R}^n$, one is led to guess that $S_m(n)$ should be a polynomial in n of degree $m+1$, which is divisible by n ($S_m(0) = 0$) and has $S_m(1) = 1$. But can we figure out which?

Bernoulli noticed that if we sum these $S_m(n)$ s over m with appropriate coefficients, we could interchange the sums and perhaps simplify. Let's try it with binomial coefficients, in an attempt to get a power of a binomial:

$$\begin{aligned} \sum_{k=0}^m \binom{m}{k} S_k(n) &= \binom{m}{0} (1^0 + 2^0 + \cdots + n^0) \\ &\quad + \binom{m}{1} (1^1 + 2^1 + \cdots + n^1) + \cdots \\ &\quad + \binom{m}{m} (1^m + 2^m + \cdots + n^m) \\ &= \left(\binom{m}{0} 1^0 + \binom{m}{1} 1^1 + \cdots + \binom{m}{m} 1^m \right) \\ &\quad + \left(\binom{m}{0} 2^0 + \binom{m}{1} 2^1 + \cdots + \binom{m}{m} 2^m \right) + \cdots \\ &\quad + \left(\binom{m}{0} n^0 + \binom{m}{1} n^1 + \cdots + \binom{m}{m} n^m \right) \\ &= (1+1)^m + (2+1)^m + \cdots + (n+1)^m \\ &= S_m(n+1) - 1 \end{aligned}$$

Since $S_m(n+1) - S_m(n) = (n+1)^m$ (clear from the definition), we can simplify this a bit by subtracting $S_m(n)$ on both sides:

$$\sum_{k=0}^{m-1} \binom{m}{k} S_k(n) = (n+1)^m - 1$$

Great—we obtained an identity relating these sums! In particular, we can now see that if $S_k(n)$ is a degree $k+1$ polynomial in n for $1 \leq k < m-1$, then S_{m-1} is a polynomial in n of degree m , and furthermore it has leading coefficient $1/m$, so this holds for all m by induction. This also gives us an easy way to compute $S_m(n)$ recursively: for example, to

obtain the formula for sums of cubes, we compute

$$\begin{aligned} S_3(n) &= \frac{1}{4} \left((n+1)^4 - 1 - 6S_2(n) - 4S_1(n) - S_0(n) \right) \\ &= \frac{1}{4} \left((n+1)^4 - 1 - n(n+1)(2n+1) - 2n(n+1) - n \right) \\ &= \frac{n^4 + 2n^3 + n^2}{4} = \frac{n^2(n+1)^2}{4} \end{aligned}$$

This agrees with $1^3 + 2^3 + 3^3 = 36 = 3^2 \cdot 4^2/4$, but also lets us easily compute

$$1^3 + 2^3 + \cdots + 1000^3 = 250000 \cdot 1002001 = 250500250000.$$

Bernoulli realized that there was a nicer way of organizing the content in these polynomials. He defined the numbers which now bear his name:

Definition 2.1. Let $B_0 = 1$, and for any $n \geq 1$, define the n th Bernoulli number B_n inductively by

$$B_n = -\frac{1}{n+1} \sum_{k=0}^{n-1} \binom{n+1}{k} B_k.$$

These are similarly easy to recursively compute: the first few are $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_3 = 0$, $B_4 = -1/30$, $B_5 = 0$, $B_6 = 1/42$, etc.

Given the form of the above system of linear equations they satisfy, involving themselves and binomial coefficients, one might reasonably guess that their generating function satisfies some kind of nice product relation with another power series, and indeed it does:

$$\begin{aligned} \left(\sum_{i=0}^{\infty} B_i \frac{x^i}{i!} \right) \left(\sum_{j=0}^{\infty} \frac{x^j}{j!} \right) &= \sum_{k=0}^{\infty} \left(\sum_{i=0}^k \binom{k}{i} B_i \right) \frac{x^k}{k!} \\ &= 1 + \frac{x}{2} + \sum_{k=2}^{\infty} (B_k + kB_{k-1} - kB_{k-1}) \frac{x^k}{k!} \\ &= x + \sum_{k=0}^{\infty} B_k \frac{x^k}{k!} \end{aligned}$$

Simplifying, we obtain

$$\sum_{k=0}^{\infty} B_k \frac{x^k}{k!} = \frac{x}{e^x - 1}$$

It's now easy to verify in fact that $B_n = 0$ for $n \geq 3$ odd. Note that after adding $x/2$ to kill the B_1 term, the generating function is even:

$$\begin{aligned} \frac{x}{e^x - 1} + \frac{x}{2} - \left(\frac{-x}{e^{-x} - 1} + \frac{-x}{2} \right) &= \frac{x(e^{-x} - 1) + x(e^x - 1)}{2 - e^x - e^{-x}} + x \\ &= \frac{x(e^{-x} + e^x - 2)}{2 - e^x - e^{-x}} + x \\ &= -x + x = 0 \end{aligned}$$

As another remark to tie things together, the connection with the power sums $S_m(n)$ from earlier is the following:

$$S_m(n) = \sum_{k=0}^m \binom{m}{k} \frac{B_k}{m+1-k} (n+1)^{m+1-k}$$

Proof. Consider $1 + e^x + e^2x + \cdots + e^{nx}$: on one hand, it's the geometric series

$$\begin{aligned} 1 + e^x + e^2x + \cdots + e^{nx} &= \frac{1 - e^{(n+1)x}}{1 - e^x} \\ &= \frac{e^{(n+1)x} - 1}{x} \frac{x}{e^x - 1} \\ &= \left(\sum_{i=0}^{\infty} \frac{(n+1)^{i+1}}{i+1} \frac{x^i}{i!} \right) \left(\sum_{j=0}^{\infty} B_j \frac{x^j}{j!} \right) \end{aligned}$$

On the other hand, it's

$$1 + e^x + e^2x + \cdots + e^{nx} = \sum_{m=0}^{\infty} (1^m + 2^m + \cdots + n^m) \frac{x^m}{m!} = \sum_{m=0}^{\infty} S_m(n) \frac{x^m}{m!}$$

Comparing m th coefficients yields the desired result. $\square$

We've now seen the definition and history of Bernoulli numbers, and one use for them. But to solve our second problem, we need another number-theoretic technology.

3 Evaluating the zeta function

Recall the definition of the *Riemann zeta function* $\zeta(s)$:

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}$$

We discussed $\zeta(s)$ at some length in class. The above series converges for any s with $\operatorname{Re}(s) > 1$, and uniformly so for $\operatorname{Re}(s) \geq 1 + \epsilon$, for any $\epsilon > 0$. We showed that it admits an analytic continuation to the complex plane without 1, with a simple pole at $s = 1$, but we won't need that here.

We also defined an analogous zeta function $\zeta_K(s)$ for any number field K , and we saw that the residue of its pole at $s = 1$ is given by the celebrated *analytic class number formula*:

$$\lim_{s \rightarrow 1} \zeta_K(s)(s-1) = \frac{2^{r_1} (2\pi)^{r_2} h_K R_K}{w_K \sqrt{|D_K|}}$$

where (r_1, r_2) is the signature of K , h_K is the class number of K , R_K is the regulator of K , w_K is the number of roots of unity in K , and D_K is the discriminant of K . Evidently, zeta

functions must encode some deep arithmetic information - so one might reasonably be interested in evaluating $\zeta(s)$ at some points.

Euler came up with a remarkable computation allowing us to do just that: we will compute $\zeta(2k)$ for all integers $k \geq 1$. The story goes as follows. Perhaps you can accept as a fact from analysis the infinite product expansion

$$\sin(x) = x \prod_{n \geq 1} \left(1 - \left(\frac{x}{n\pi} \right)^2 \right).$$

This makes sense: $\sin(x)$ has zeros precisely at the integer multiples of π . Let's first take logarithms:

$$\begin{aligned} \log \sin(x) &= \log(x) + \sum_{n \geq 1} \log \left(1 - \left(\frac{x}{n\pi} \right)^2 \right) \\ &= \log(x) - \sum_{n \geq 1} \sum_{m \geq 1} \frac{1}{m} \left(\frac{x}{n\pi} \right)^{2m} \end{aligned}$$

then differentiate:

$$\frac{\cos(x)}{\sin(x)} = \frac{1}{x} - 2 \sum_{n \geq 1} \sum_{m \geq 1} \frac{1}{n\pi} \left(\frac{x}{n\pi} \right)^{2m-1}$$

Multiplying by x and swapping the order of summation, we have

$$\begin{aligned} x \frac{\cos(x)}{\sin(x)} &= 1 - 2 \sum_{n \geq 1} \sum_{m \geq 1} \left(\frac{x}{n\pi} \right)^{2m} \\ &= 1 - 2 \sum_{m \geq 1} \left(\frac{x}{\pi} \right)^{2m} \sum_{n \geq 1} \frac{1}{n^{2m}} \\ &= 1 - 2 \sum_{m \geq 1} \left(\frac{x}{\pi} \right)^{2m} \zeta(2m) \end{aligned}$$

But on the other hand, rewriting this using the exponential identities for $\sin(x)$ and $\cos(x)$, we have

$$\begin{aligned} x \frac{\cos(x)}{\sin(x)} &= ix \frac{e^{ix} + e^{-ix}}{e^{ix} - e^{-ix}} \\ &= ix \frac{e^{2ix} - 1 + 2}{e^{2ix} - 1} \\ &= ix + \frac{2ix}{e^{2ix} - 1} \\ &= ix + \sum_{k \geq 0} B_k \frac{(2ix)^k}{k!} \end{aligned}$$

Now we can carefully compare coefficients of these two power series: we obtain

$$\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{2 \cdot (2m)!} B_{2m}$$

concluding the calculation.

Let's see some examples. For $m = 1$, we recover the classical Basel problem:

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \pi^2 B_2 = \frac{\pi^2}{6}.$$

But we could perform further computations, e.g.

$$\zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}, \quad \zeta(6) = \frac{\pi^6}{945}.$$

One might reasonably ask: sure, these are cool computations. But do they answer any other concrete arithmetic questions? In fact, they provide the answer to the second problem posed in the introduction: calculating the density of squarefree (or $2m$ -free) numbers. A number n is k -free if and only if $n \not\equiv 0 \pmod{p^k}$ for every prime p . By the Chinese Remainder Theorem (applied to any finite subset of primes), the proportion of residue classes (hence the proportion of integers) satisfying this is multiplicative in p , that is, the product over such p of $(1 - 1/p^k)$. In the limit as we consider all primes, we see that the density of k -free numbers is

$$\prod_{\text{primes } p} \left(1 - \frac{1}{p^k}\right)$$

But recall the well-known Euler product expansion for $\zeta(s)$:

$$\zeta(s) = \prod_{\text{primes } p} \left(1 - \frac{1}{p^s}\right)^{-1}$$

Thus, the density of k -free numbers is simply $1/\zeta(k)$. In the k even case, we have computed these: the density of $2m$ -free integers is

$$\frac{1}{\zeta(2m)} = (-1)^{m+1} \frac{2 \cdot (2m)!}{(2\pi)^{2m} B_{2m}}$$

As concrete examples, we obtain that $6/\pi^2 \approx 60.8\%$ of integers are square-free, $90/\pi^4 \approx 92.4\%$ of integers are 4th-power-free, and $945/\pi^6 \approx 98.3\%$ of integers are 6th-power-free.